LESSON QUIZZES

Answers

Short answers for all 17 quizzes. The written question shows the final answer.

LESSON 11-1 · ADDING AND SUBTRACTING POLYNOMIALS

Q1 D **Q2** B **Q3** 6 **Q4** B

Q5 B **Q6** B

Q7 Degrees: 7 → 0, $-2x^2$ → 2, $9x^4$ → 4, $-x$ → 1. · Standard form: $9x^4 - 2x^2 - x + 7$. · Degree 4 (quartic), leading coefficient 9. · 4 terms → a polynomial (not mono-, bi-, or trinomial).

LESSON 11-2 · MULTIPLYING POLYNOMIALS BY MONOMIALS

Q1 D **Q2** B **Q3** 4 **Q4** A

Q5 A **Q6** D

Q7 $-20d^3 + 48d + 7d + 35$ · $= -20d^3 + 55d + 35$

LESSON 11-3 · MULTIPLYING POLYNOMIALS

Q1 D **Q2** A **Q3** D **Q4** A

Q5 D **Q6** D **Q7** B **Q8** D

LESSON 11-4 · SPECIAL PRODUCTS

Q1 B **Q2** A **Q3** A **Q4** D

Q5 D **Q6** D

Q7 (a) $25y^2 + 40y + 16$ · (b) $4a^2 - 12a + 9$ · (c) $x^4 - 9y^2$

LESSON 11-5 · USING THE DISTRIBUTIVE PROPERTY

Q1 D **Q2** D **Q3** A **Q4** B

Q5 B **Q6** C

Q7 GCF: numbers 3, x^2, y → $3x^2y$ · $3x^2y(2x - 3y + 4)$ · Check: $3x^2y \cdot 2x = 6x^3y$, $3x^2y \cdot (-3y) = -9x^2y^2$, $3x^2y \cdot 4 = 12x^2y$ ✓

LESSON 11-6 · FACTORING QUADRATIC TRINOMIALS

Q1 D **Q2** A **Q3** D **Q4** A

Q5 C **Q6** D

Q7 $x^2 - 6x + 9 = (x - 3)^2$, so each side is $x - 3$. · New length $x + 1$, width $x - 3$. · Area $= (x + 1)(x - 3) = x^2 - 2x - 3$ ft²

LESSON 11-7 · FACTORING SPECIAL PRODUCTS

Q1 D **Q2** A **Q3** B **Q4** D

Q5 A **Q6** C

Q7 Carpet $= 36 - x^2$ · $= 6^2 - x^2 = (6 + x)(6 - x)$ m²

LESSON 12-1 · GRAPHING QUADRATIC FUNCTIONS

Q1 C **Q2** 5 **Q3** B **Q4** D

Q5 76 **Q6** 400

Q7 (a) $t = -\frac{32}{2(-16)} = 1$ s · (b) $h(1) = -16 + 32 + 48 = 64$ ft · (c) $h(0) = 48$ ft

LESSON 12-2 · TRANSFORMATIONS OF QUADRATIC FUNCTIONS

Q1 D **Q2** A **Q3** B **Q4** A

Q5 D **Q6** D

Q7 $a = -\frac{1}{2}$: reflected across the **x**-axis and compressed vertically. · **h = −4**: 4 units left · **k = −3**: 3 units down. · Vertex **(−4, −3)**; opens down → maximum value −3.

LESSON 12-3 · SOLVING QUADRATIC EQUATIONS BY GRAPHING

Q1 A **Q2** C **Q3** A **Q4** B

Q5 0.5 **Q6** 1.3

Q7 Axis **x = 2**, vertex **(2, 9)**, opens down · Table: **x** = −1, 0, 2, 4, 5 → **y** = 0, 5, 9, 5, 0 · Solutions: **x = −1** and **x = 5**

LESSON 12-4 · SOLVING QUADRATIC EQUATIONS BY FACTORING

Q1 D **Q2** C **Q3** D **Q4** B

Q5 A **Q6** D

Q7 $3b = 0 \rightarrow b = 0$; $9b - 27 = 0 \rightarrow b = 3$ · $a^2 - 4a = 0 \rightarrow a(a - 4) = 0 \rightarrow a = 0$ or $a = 4$ · Check by substituting each value ✓

LESSON 12-5 · SOLVING QUADRATIC EQUATIONS BY COMPLETING THE SQUARE

Q1 C **Q2** D **Q3** B **Q4** B

Q5 B **Q6** B

Q7 $x^2 + 6x = 16$ · $x^2 + 6x + 9 = 25$ · $(x + 3)^2 = 25 \rightarrow x + 3 = \pm 5$ · $x = 2$ or $x = -8$

LESSON 12-6 · SOLVING QUADRATIC EQUATIONS BY USING THE QUADRATIC FORMULA

Q1 B **Q2** A **Q3** D **Q4** D

Q5 A **Q6** A

Q7 $0.01t^2 + 0.5t - 13 = 0$ · $t = \frac{-0.5 \pm \sqrt{0.77}}{0.02}$ · $t \approx 18.9$ (reject -68.9)

LESSON 12-7 · SOLVING SYSTEMS OF LINEAR AND QUADRATIC EQUATIONS

Q1 C **Q2** A **Q3** A **Q4** D

Q5 D **Q6** A **Q7** A **Q8** C

LESSON 10-1 · MULTIPLICATION PROPERTIES OF EXPONENTS

Q1 A **Q2** A **Q3** C **Q4** 14

Q5 D **Q6** A

Q7 $11x^{6+1}y^{8+9} = 11x^7y^{17}$ · $8m^{3+1}n^{1+5} = 8m^4n^6$

LESSON 10-2 · DIVISION PROPERTIES OF EXPONENTS

Q1 D **Q2** A **Q3** C **Q4** D

Q5 D **Q6** C

Q7 a^6b^3 **d** (combine $a^2 \cdot a^4$, subtract the others) · Inside: $4mn^2p$ **3** · $256m^4n^8p^4$ **81**

LESSON 10-3 · NEGATIVE EXPONENTS

Q1 B **Q2** B **Q3** D **Q4** C

Q5 C **Q6** A

Q7 (a) $-\frac{1}{12}g^{-6}h^6 = -\frac{h^6}{12g^6}$ · (b) $\frac{5}{2}r^{-1}s^{-3} = \frac{5}{2rs^3}$